



LAB 1

Computer Forensics



JANUARY 17, 2025

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Student ID: 100997946

Sometimes discovery demands from law firms require you to recover only allocated data from a disk.

Note: You will be graded on the screenshot you attach of every step for proof of work.

Download all the files attached with this document to analyze it.

Task 1: Show how you can extract just the files that haven't been deleted (that is, the allocated files) from an image.

1. Start Autopsy for Windows. Click the Create New Case icon. In the New Case Information window, enter C1Prj04 in the Case Name text box, and click Browse next to the Base Directory text box. Navigate to and click your work folder, and then click Next.

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name: C1Prj04

Base Directory: E:\Durham_College\Sem_2\Computer_Foring\Lab_1 Browse

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

E:\Durham_College\Sem_2\Computer_Foring\Lab_1\C1Prj04

< Back **Next >** Finish Cancel Help

2. In the Additional Information window, type C1Prj04 in the Case Number text box and your name in the Examiner text box, and then click Finish.

New Case Information

Steps

1. Case Information
2. Optional Information

Optional Information

Case

Number: C1Prj04

Examiner

Name: Madhvi Lashkaree

Phone: 9054248182

Email: madhvi.lashkaree@dcmail.ca

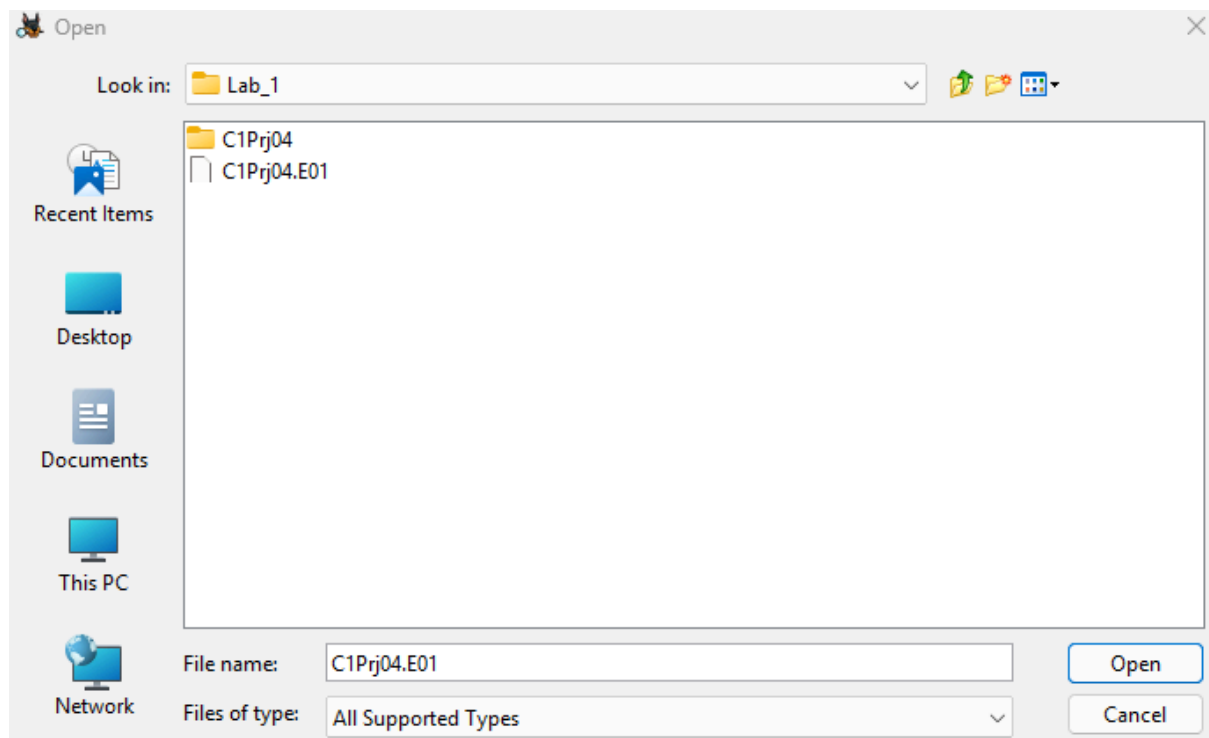
Notes:

Organization

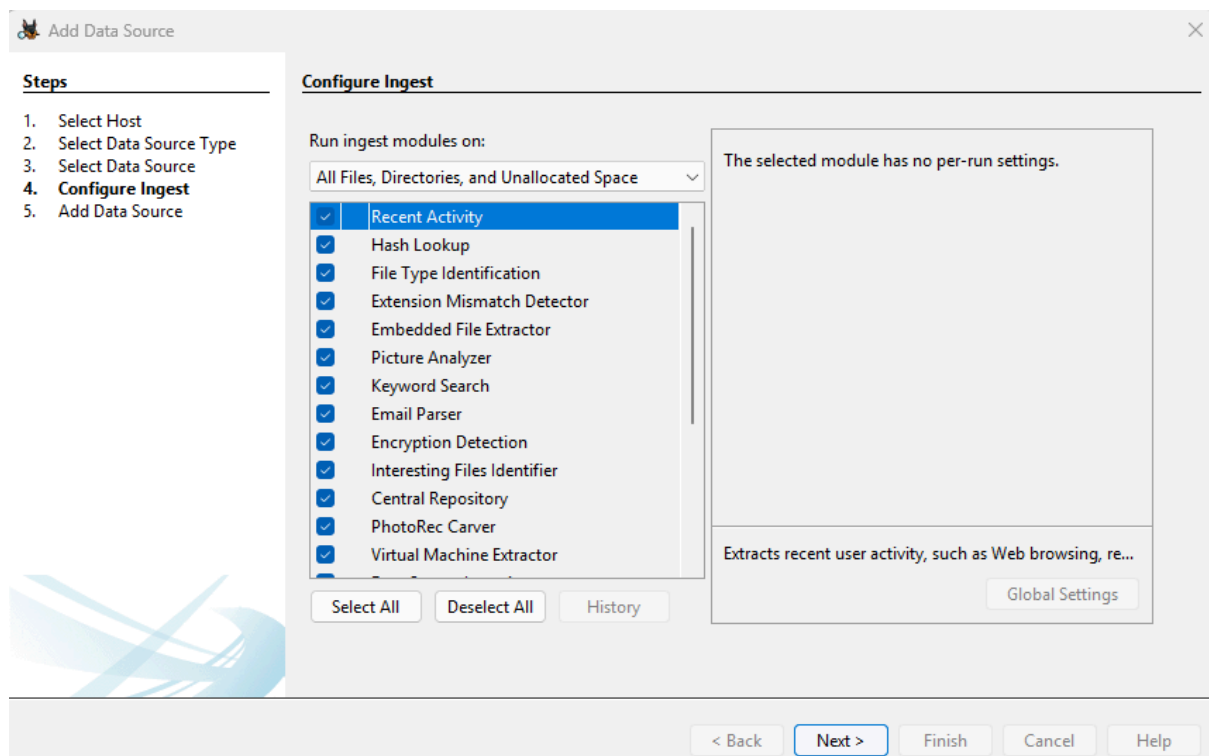
Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > **Finish** Cancel Help

3. In the Select Data Source window, click the Select data source type list arrow, and click Disk Image or VM file. Click the Browse button next to the “Browse for an image” text box, navigate to your work folder and click the C1Prj04.E01 file, and then click Open. Click Next.



4. In the Configure Ingest Modules dialog box, click Select All. Click Next and then Finish.



- In the Tree Viewer pane, expand Views, File Types, and By Extension. Under By Extension are several subfolders representing file types, as you've seen in previous projects. Next to each file type subfolder is a number enclosed in parentheses, which shows the number of files of this type Autopsy found. Click subfolders with numbers greater than zero to view the files.

The screenshot shows the Autopsy 4.21.0 interface. The Tree Viewer pane on the left is expanded to 'By Extension' > 'Images (5)'. The right pane shows a 'Listing' of images with columns Name, S, C, and O.

Name	S	C	O
Gettysburg.jpg			
Lin_tomb.jpg			0
Lincoln.jpg			
Gettysbg.jpg			0
f0000000.jpg			0

- In the Result Viewer pane, scroll to the right, if necessary until the Flags(Meta) column is in view. Sort the column by clicking the Flags(Meta) header, which displays all allocated files to the top of the list.

Note: In the Result Viewer pane, allocated (not deleted) files have a paper sheet icon to the left of the filename. Deleted (unallocated) files have a red X over the paper sheet icon, and unallocated files that have been corrupted and recovered by the OS have a diagonal broken bar icon.

Listing

Images

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
Lin_tomb.jpg			1	2004-06-23 21:29:56 EDT	0000-00-00 00:00:00	2006-09-24 00:00:00 EDT	2004-06-23 22:36:08 EDT	9950	Allocated	Allocated	unknown	/img_C1Prj04.E01/Lin_tomb.jpg
Gettysbg.jpg			1	2004-06-23 21:20:52 EDT	0000-00-00 00:00:00	2006-09-24 00:00:00 EDT	2004-06-23 22:36:05 EDT	58927	Allocated	Allocated	unknown	/img_C1Prj04.E01/Gettysbg.jpg
Gettysburg.jpg				2004-06-23 21:20:52 EDT	0000-00-00 00:00:00	2004-06-23 00:00:00 EDT	2004-06-23 22:36:05 EDT	58927	Unallocated	Unallocated	unknown	/img_C1Prj04.E01/Gettysburg.jpg
Lincoln.jpg				2004-06-23 21:29:36 EDT	0000-00-00 00:00:00	2004-06-23 00:00:00 EDT	2004-06-23 22:36:12 EDT	18553	Unallocated	Unallocated	unknown	/img_C1Prj04.E01/Lincoln.jpg
f0000000.jpg			1	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	18553	Unallocated	Unallocated	unknown	/img_C1Prj04.E01/ScarvedFiles/1/f0000000

7. Scroll to the left until the Name column is visible. If there are allocated files, they will be at the top of this list. Ctrl+click each allocated file, right-click this selection, and then click Extract File(s).

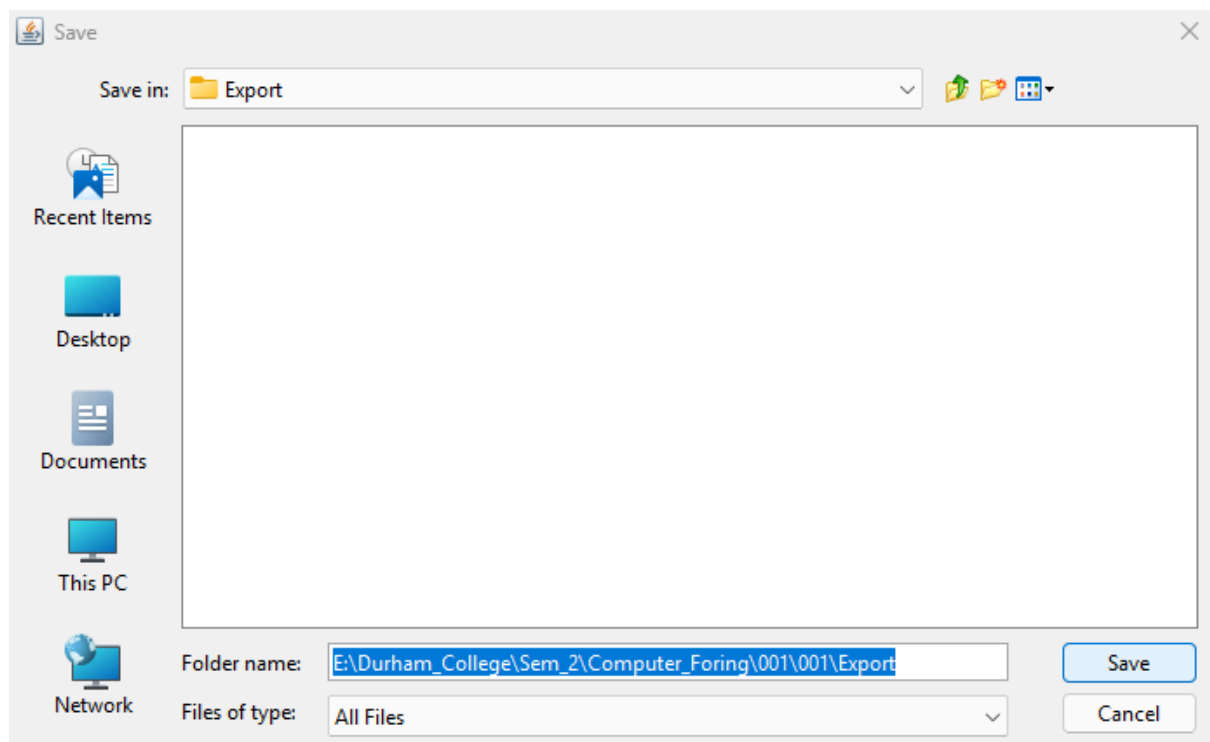
Listing

Images

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)
Lin_tomb.jpg			1	2004-06-23 21:29:56 EDT	0000-00-00 00:00:00	2006-09-24 00:00:00 EDT	2004-06-23 22:36:08 EDT	9950	Allocated	Allocated
Gettysbg.jpg				EDT	0000-00-00 00:00:00	2006-09-24 00:00:00 EDT	2004-06-23 22:36:05 EDT	58927	Allocated	Allocated
Gettysburg.jpg				EDT	0000-00-00 00:00:00	2004-06-23 00:00:00 EDT	2004-06-23 22:36:05 EDT	58927	Unallocated	Unallocated
Lincoln.jpg				EDT	0000-00-00 00:00:00	2004-06-23 00:00:00 EDT	2004-06-23 22:36:12 EDT	18553	Unallocated	Unallocated
f0000000.jpg					0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	18553	Unallocated	Unallocated

8. In the Save dialog box, click Save to save the files automatically in Autopsy's case subfolder:
E:\Durham_College\Sem_2\Computer_Foring\001\001\Export.



9. Write a brief memo that lists all the files you exported. Leave Autopsy running for the next project.

Date: January 17, 2025

Subject: Exported Files from Autopsy Lab 1 Task 1

Location: E:\Durham_College\Sem_2\Computer_Foring\001\001\Export

The following allocated files were successfully exported from the disk image *C1Prj04.E01* using Autopsy's file extraction feature, adhering to standard forensic procedures for data recovery. The extracted files include two historical images: one depicting a memorial monument with an obelisk and statues, and another containing the manuscript of the *Gettysburg Address*. These files were recovered as allocated (non-deleted) data, ensuring their integrity and authenticity for further analysis.



ENGLISH

ADDRESS DELIVERED AT THE DEDICATION
OF THE CEMETERY AT GETTYSBURG

Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal.

Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure. We are met on a great battle-field of that war. We have come to dedicate a portion of that field, as a final resting place for those who here gave their lives that that nation might live. It is altogether fitting and proper that we should do this.

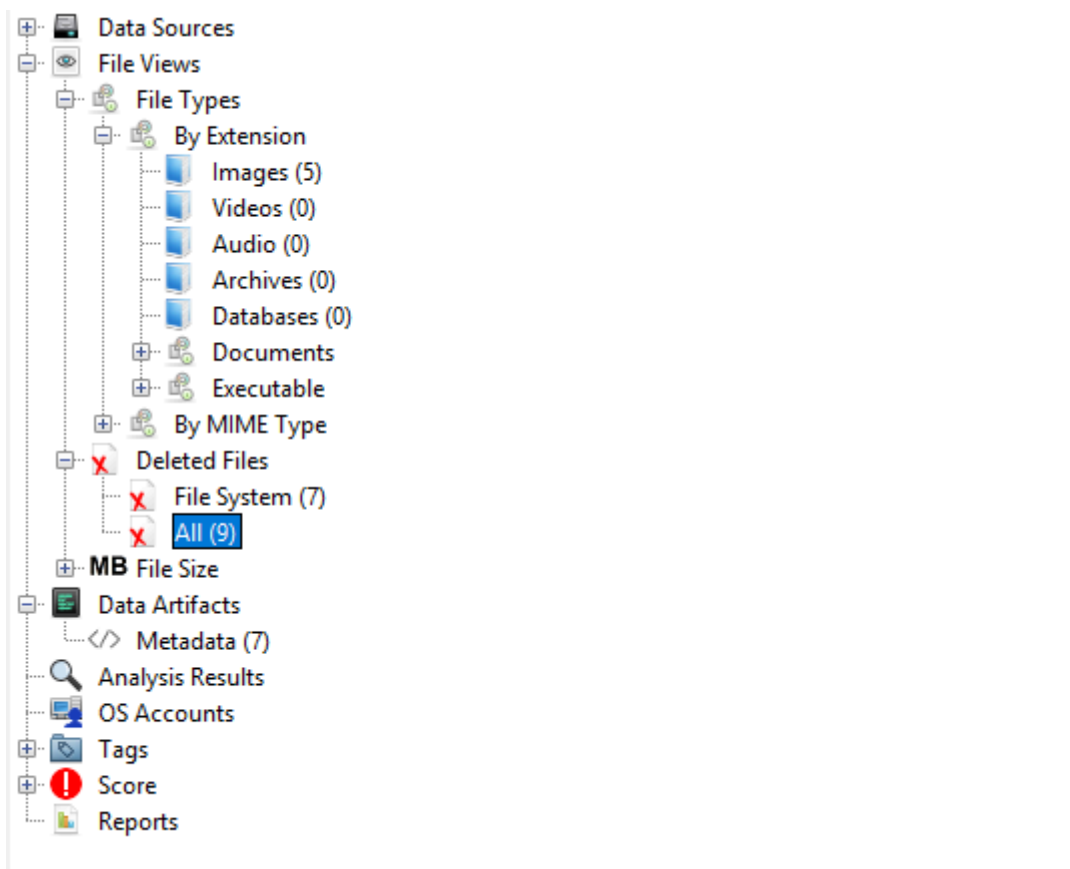
But, in a larger sense, we can not dedicate—we can not consecrate—we can not hallow—this ground. The brave men, living and dead, who struggled here, have consecrated it, far above our poor power to add or detract. The world will little note, nor long remember what we say here, but it can never forget what they did here. It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us—that from these honored dead we take increased devotion to that cause for which they gave the last full measure of devotion—that we here highly resolve that these dead shall not have died in vain—that this nation, under God, shall have a new birth of freedom—and that government of the people, by the people, for the people, shall not perish from the earth.

ABRAHAM LINCOLN.

November 19, 1863.

Task 2: You create a report listing all the unallocated (deleted) files Autopsy finds.

1. Start Autopsy for Windows and click the Open Recent Case icon, if necessary.
2. In the Tree Viewer pane, expand Views, File Types, Deleted Files, and All.



3. In the Result Viewer pane, Ctrl+click all files in the All subfolder. Right-click this selection, point to Tag File and then Quick Tag, and click Follow Up.

The screenshot shows the 'Data Artifacts' tab in the application. A table is visible with columns: S, C, O, Modified Time, Change Time, and Access Time. The table contains several rows of data, including file names like 'ENCE.doc', 'ON.doc', and 'loc'. A context menu is open over the table, with the 'Add File Tags' option selected. This opens a sub-menu where 'Follow Up' is highlighted. Other options in the sub-menu include 'Bookmark', 'Notable Item (Notable)', 'Project VIC', 'Tag and Comment...', and 'New Tag...'. The 'Properties' option in the main context menu is also visible.

- Click Generate Report at the top. In the Generate Report window, click the Results - Excel option button in the Report Modules section, and then click Next.

The screenshot shows the 'Generate Report' window with the title bar 'Generate Report' and a close button. The main section is titled 'Select and Configure Report Modules'. On the left, under 'Report Modules:', there is a list of radio buttons: 'HTML Report', 'Excel Report' (which is selected), 'Files - Text', 'Data Source Summary Report', 'Save Tagged Hashes', 'Extract Unique Words', 'TSK Body File', 'Google Earth KML', 'CASE-UCO', and 'Portable Case'. To the right of this list, there is a text area that says 'A report about results and tagged items in Excel (XLS) format.' and a larger text box containing the message 'This report will be configured on the next screen.' At the bottom of the window, there are five buttons: '< Back', 'Next >' (highlighted with a blue border), 'Finish', 'Cancel', and 'Help'.

- In the Configure Artifacts Report window, click the Tagged Results button, click the Follow Up check box, and then click Finish.

The screenshot shows the 'Generate Report' window with the title bar 'Generate Report' and a close button. The main section is titled 'Configure Report'. Under the heading 'Select which data to report on:', there are three radio buttons: 'All Results', 'All Tagged Results' (which is selected), and 'Specific Tagged Results'. Below this, there is a large text box containing a checked checkbox labeled 'Follow Up'. To the right of this text box are two buttons: 'Select All' and 'Deselect All'. At the bottom left of the text box area, there is a button labeled 'Choose Result Types...'. At the bottom of the window, there are five buttons: '< Back', 'Next >', 'Finish' (highlighted with a blue border), 'Cancel', and 'Help'.

6. In the Report Generation Progress Complete window, click the Results - Excel pathname to open the Excel report. When you're finished, click Close in the Report Generation Progress window.

Listing			
Table Thumbnail Summary			
Source Module Name	Report Name	Created Time	Report File Path
 Excel Report		2025-01-17 12:35:21 EST	E:\Durham_College\Sem_2\Computer_Foring\001\001\Reports\001 E
 Excel Report		2025-01-17 12:42:14 EST	E:\Durham_College\Sem_2\Computer_Foring\001\001\Reports\001 E

7. Write a brief memo listing all the deleted files on your report spreadsheet. Exit Autopsy.

Date: January 17, 2025

Subject: Deleted Files Report from Case 001

Examiner: Madhvi Lashkaree

During the forensic analysis of *C1Prj04.E01*, the following deleted files were identified and recovered from the carved files directory using forensic carving techniques:

- File Path:** */img_C1Prj04.E01/\$CarvedFiles/1/f0000000.jpg*
 - Size:** 18,553 bytes
 - Hash:** c2d255934b7613177e47a580ee51c1e0
 - Timestamps:** All timestamps recorded as *0000-00-00 00:00:00*
- File Path:** */img_C1Prj04.E01/\$CarvedFiles/1/f0000037_Amendments_to_the_Constitution.doc*
 - Size:** 87,040 bytes
 - Hash:** a57ff86d6c2226031cf7813341199e82
 - Timestamps:** All timestamps recorded as *0000-00-00 00:00:00*

These files were recovered using Autopsy's data carving functionality, as their metadata had been removed or overwritten. Further analysis may be required to determine their relevance to the investigation.

Lab 4

Student Name: Madhvi (100997946)

Note: You will be graded on the screenshot you attach of every step for proof of work. Create a *Work\Chap07\Projects* folder on your system before starting the projects; it's referred to as your "work folder" in steps. All the data files you use in these projects are from previous chapters.

The purpose of this project is to become more familiar with the Linux version of Sleuth Kit and Autopsy. The best way to learn a tool, especially one that isn't well documented, is to explore its functions. You're encouraged to work in teams for this project and share your findings with other students. In this project, you use the image file **GCFI-datacarve-FAT.dd** from Lab 2 and analyze it with Sleuth Kit and Autopsy. You need a system with a Linux virtual machine and Sleuth Kit and Autopsy installed. Follow these steps:

1. Copy the **GCFI-datacarve-FAT.dd** file to your Linux virtual machine with Sleuth Kit and Autopsy installed. Start Sleuth Kit and Autopsy. In the main window, click **New Case**. In the Create a New Case dialog box, fill in your information (using GCFI-datacarve-FAT for the case name), and then click **New Case**.

CREATE A NEW CASE

1. **Case Name:** The name of this investigation. It can contain only letters, numbers, and symbols.
GCFI-datacarve-FAT

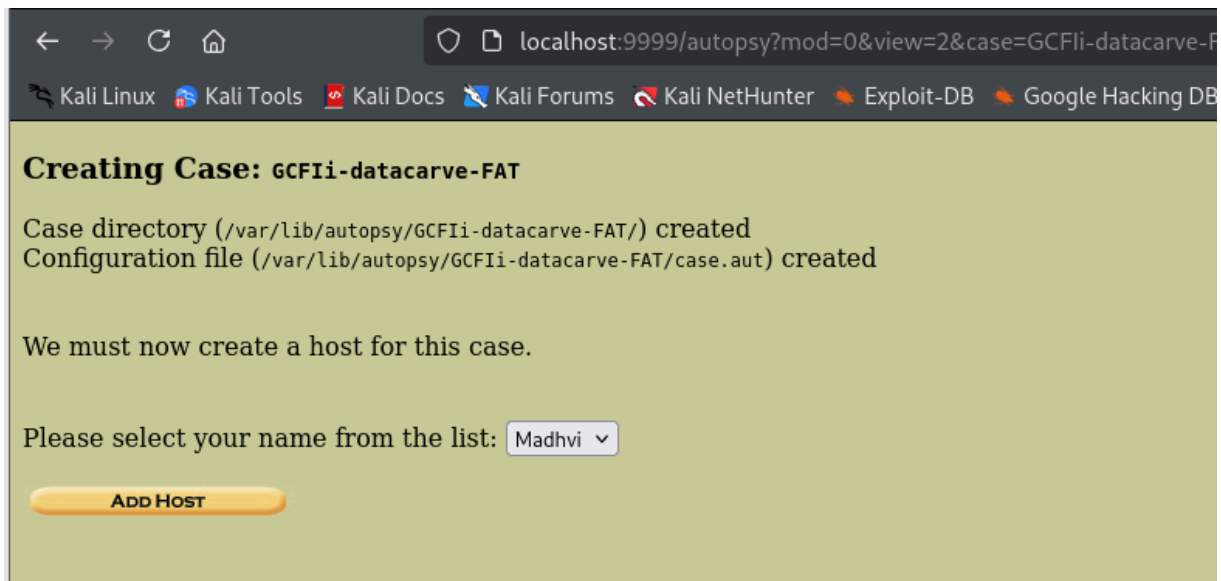
2. **Description:** An optional, one line description of this case.

3. **Investigator Names:** The optional names (with no spaces) of the investigators for this case.

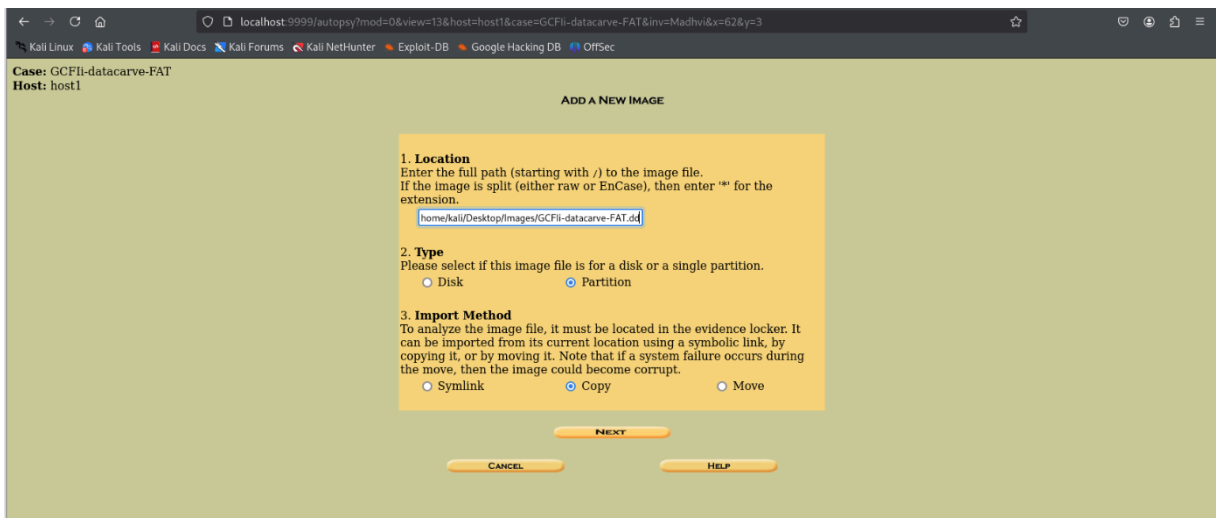
a. Madhvi	b.
c.	d.
e.	f.
g.	h.
i.	j.

NEW CASE CANCEL HELP

2. In the Creating Case dialog box, click **Add Host**, and in the Add a New Host dialog box, enter your information, and click **Add Host**.

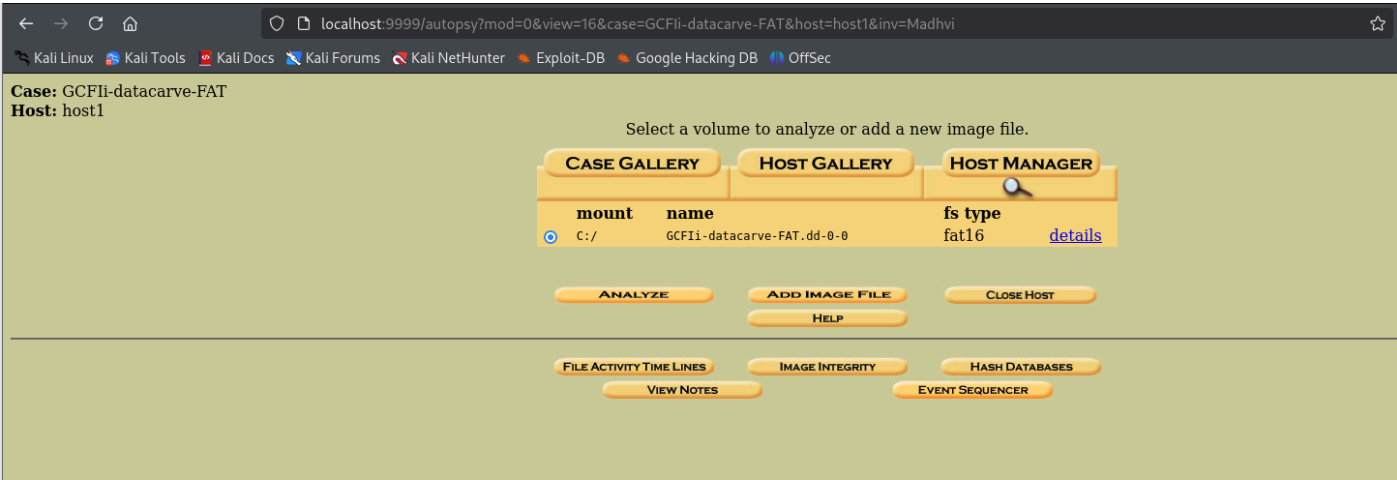
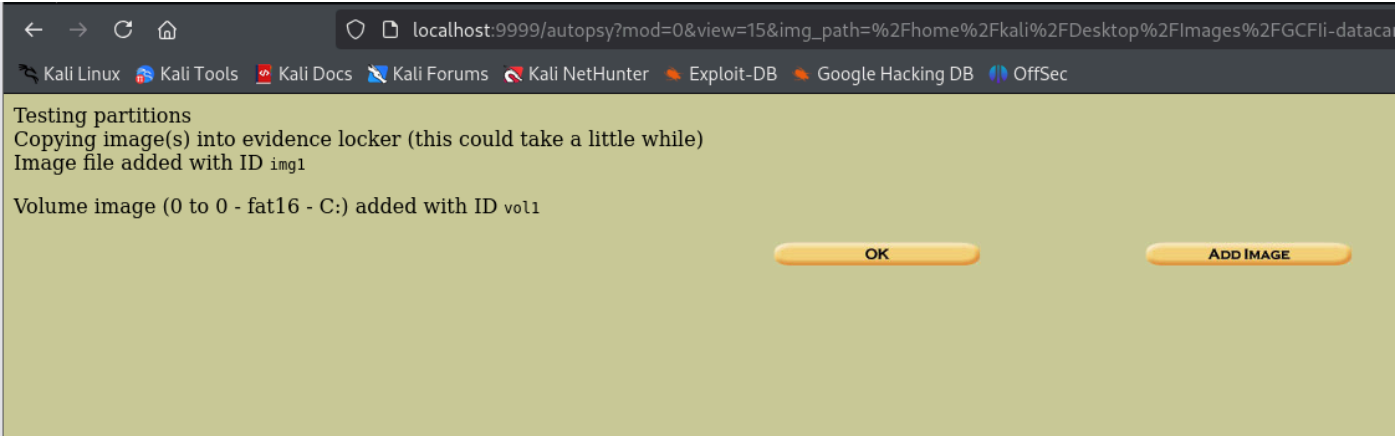


3. In the Adding host dialog box, click **Add Image** to continue. In the Open Image dialog box, click **Add Image File**. In the Add a New Image dialog box, type the full pathname and the **GCFI-datacarve-FAT.dd** image filename in the Location text box, click the **Partition** option button, click the **Copy** option button for the import method, and then click **Next**.

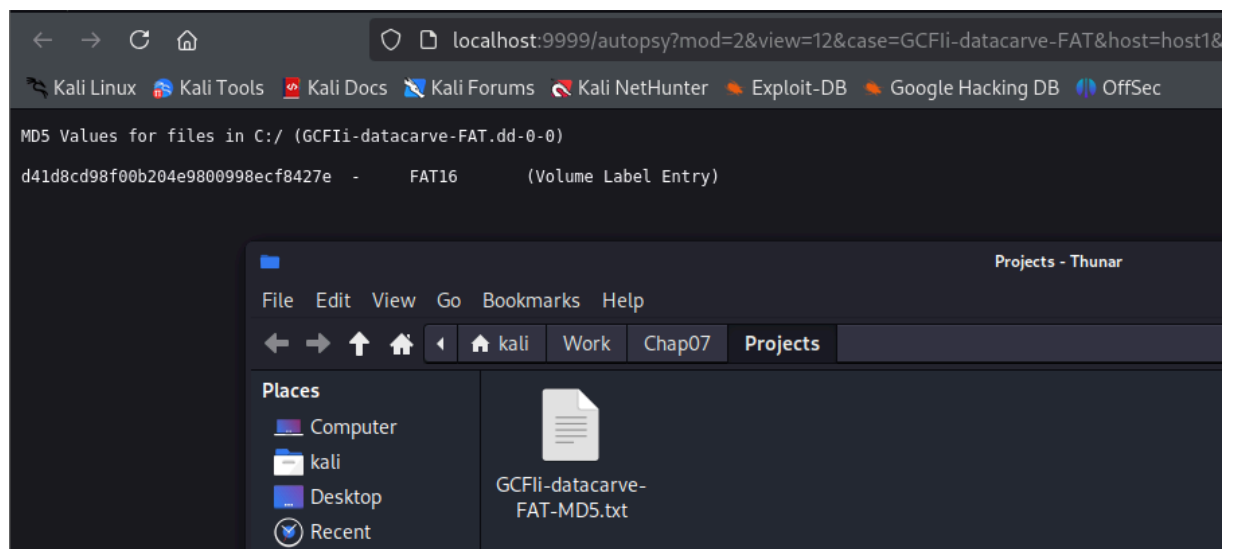
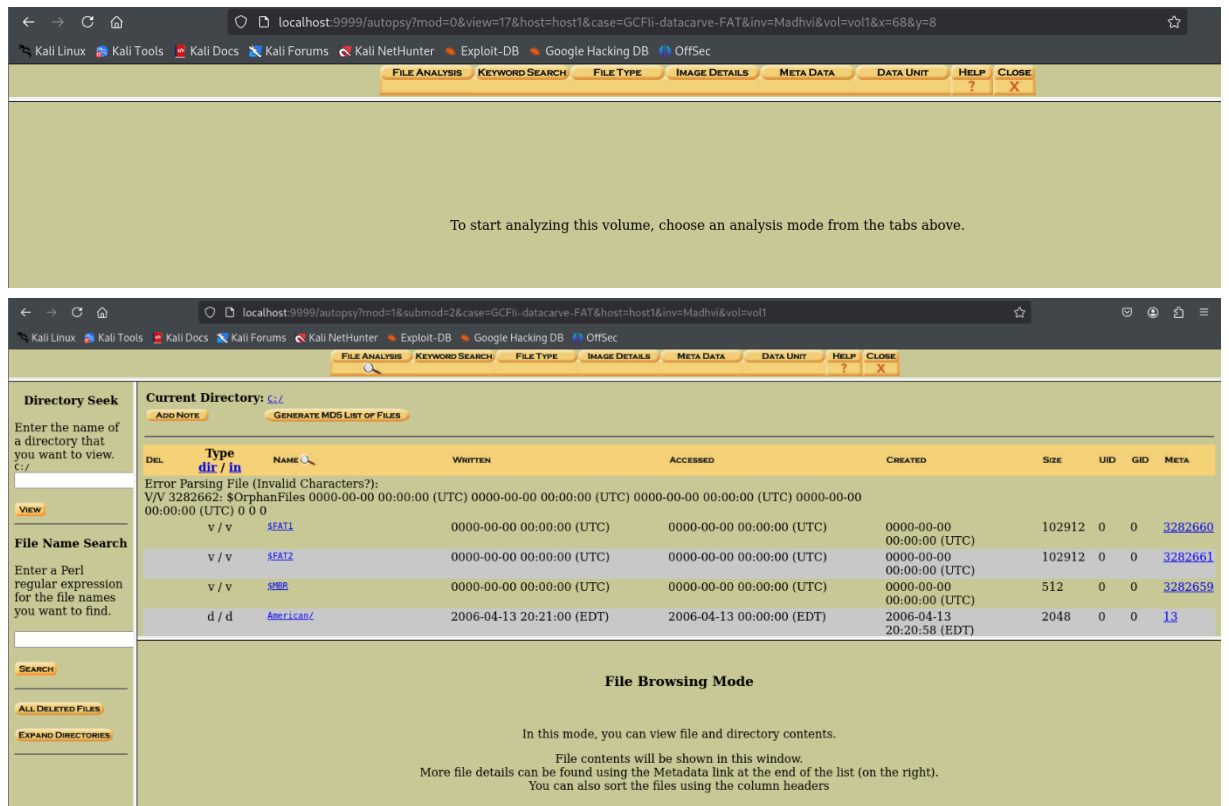


4. In the Image File Details section, click **Add**, and in the Testing partitions dialog box, click **OK**. In

the “Select a volume to analyze or add a new image file” dialog box, click the **Analyze** button.

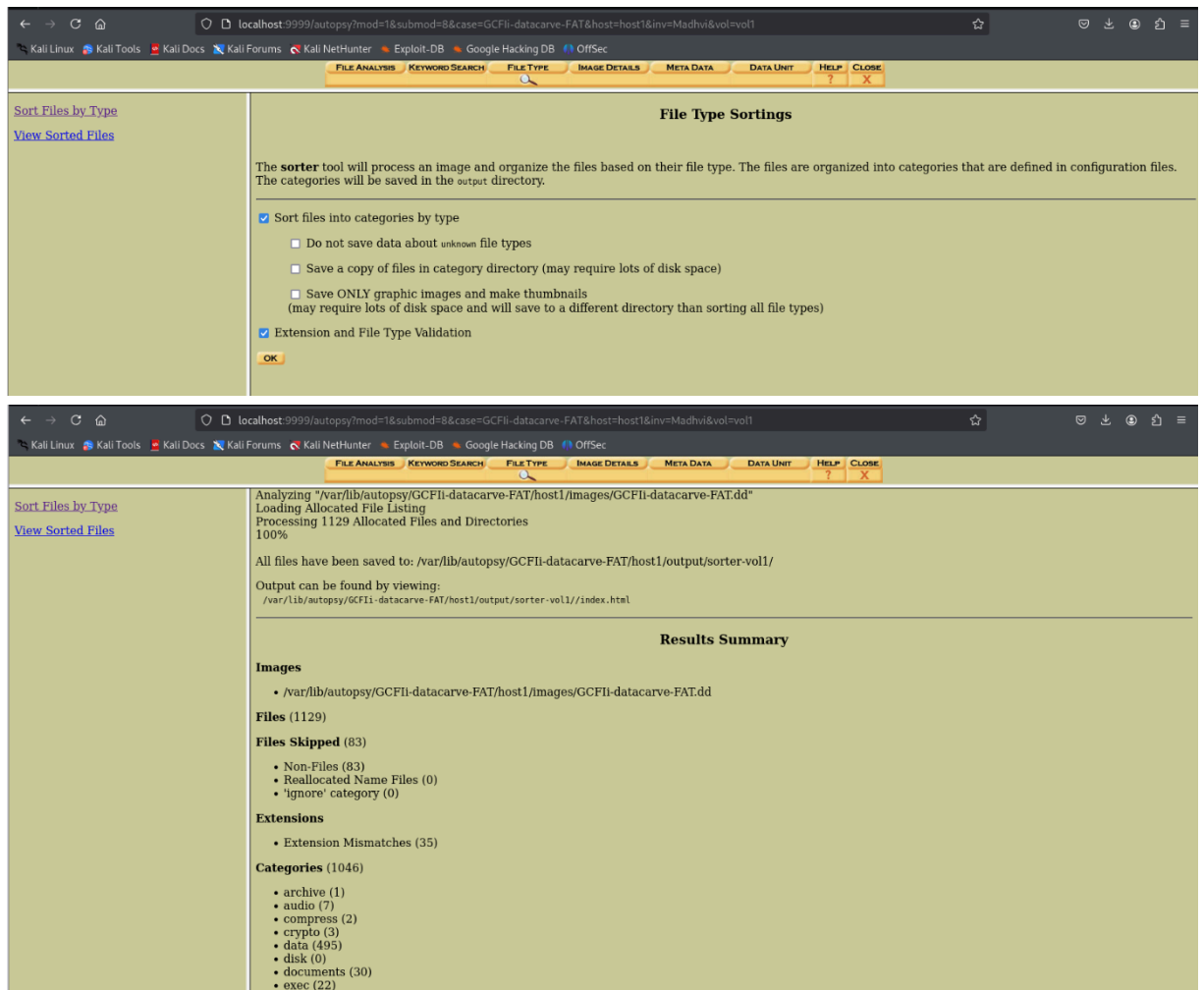


5. In the Analysis dialog box, click **File Analysis**, and then click **Generate MD5 List of Files**. In the MD5 results window, save the list as **GCFI-datacarve-FAT-MD5.txt** in your work folder, and close the MD5 results window.

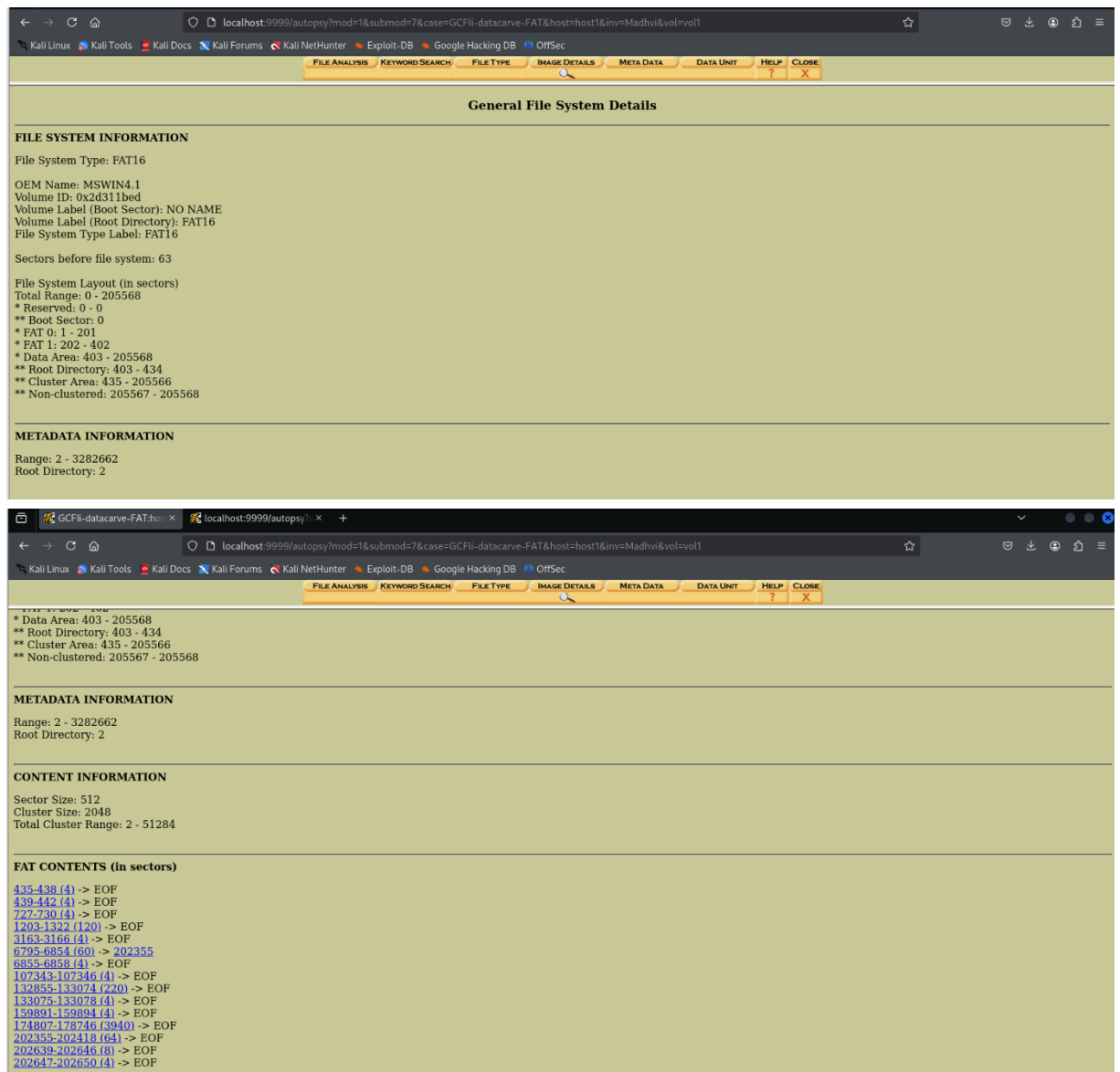


6. Next, in the Analysis dialog box, click **File Type**, click **Sort Files by Type**, and then click **OK**. When the analysis is finished, print the Results Summary frame of the

Web page.



7. Click **Image Details**, and in the General File System Details dialog box, print the frame containing the results.



- Write a report describing the information each function asks for and what information it produces so that you can begin building your own user manual for this tool.

Sleuth Kit and Autopsy Lab Report

Course/Module: Computer Forensics

Student Name: Madhvi

Date: 2025-Feb-02

Instructor: Mohamed Elghazouly

1. Introduction

This report documents the analysis of GCFIi-datacarve-FAT.dd using Sleuth Kit and Autopsy in a Kali Linux virtual environment. The primary objective of this laboratory exercise was to gain practical experience with forensic tools and understand their investigative capabilities.

2. Environment Details

The analysis was conducted using the following setup:

- **Operating System:** Kali Linux
- **Software Tools:** Sleuth Kit, Autopsy
- **Image File:** GCFIi-datacarve-FAT.dd
- **Work Folder:** /home/kali/Work/Chap07/Projects

3. Case Creation and Setup

The following parameters were used to create and configure the case:

- **Case Name:** GCFI-datacarve-FAT
- **Host Information:** host1
- **Image** **File** **Location:**
/home/user/Work/Chap07/Projects/GCFI-datacarve-FAT.dd
- **Import Method:** Copy
- **Partition Selection:** Partition option

The setup process was documented with screenshots showing the creation of a new case and the addition of host and image information.

4. File Analysis and MD5 Hash Generation

The MD5 hash generation process was completed successfully to ensure file integrity throughout the analysis. The following steps were performed:

1. Selected File Analysis in the analysis window
2. Generated MD5 List of Files
3. Saved the output as GCFI-datacarve-FATMD5.txt in the work folder

The MD5 hash values serve as digital fingerprints, verifying that no files were altered during the analysis process.

5. Sorting Files by Type

File type analysis was conducted to categorize and identify key file types present in the image. The process involved:

1. Selecting File Type in the Analysis dialog
2. Executing Sort Files by Type command

This categorization provides a systematic overview of the file system contents, facilitating targeted analysis of specific file types.

6. Image File System Details

The General File System Details examination revealed comprehensive information about:

- FAT file system structure
- Volume information
- Partition details

This information provides crucial context for understanding the storage media's organization and characteristics.

7. Conclusion

The practical experience gained through this laboratory exercise with Sleuth Kit and Autopsy has demonstrated the essential capabilities of these forensic tools. Through the analysis of GCFIi-datacarve-FAT.dd, valuable insights were gained into:

- Case creation and management
- MD5 hash generation for file integrity verification
- File type categorization and analysis
- File system structure examination

These skills and knowledge will be instrumental in conducting thorough and methodical digital forensics investigations in the future.